DATA WAREHOUSING AND DATA MINING

Jury 2 - Three Dashboard Evidence

Integrated Clinical & Operational Healthcare Intelligence Platform: Data Warehouse with AI-Driven Patient Readmission & Bed Capacity Prediction

Jury 2 focus: Predictive Patient Risk Mining and Clinical Feature Profiling using Healthcare Data

Team member name	Registration number
_____	_____
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_____	_____

Department: _____

Institution: _____

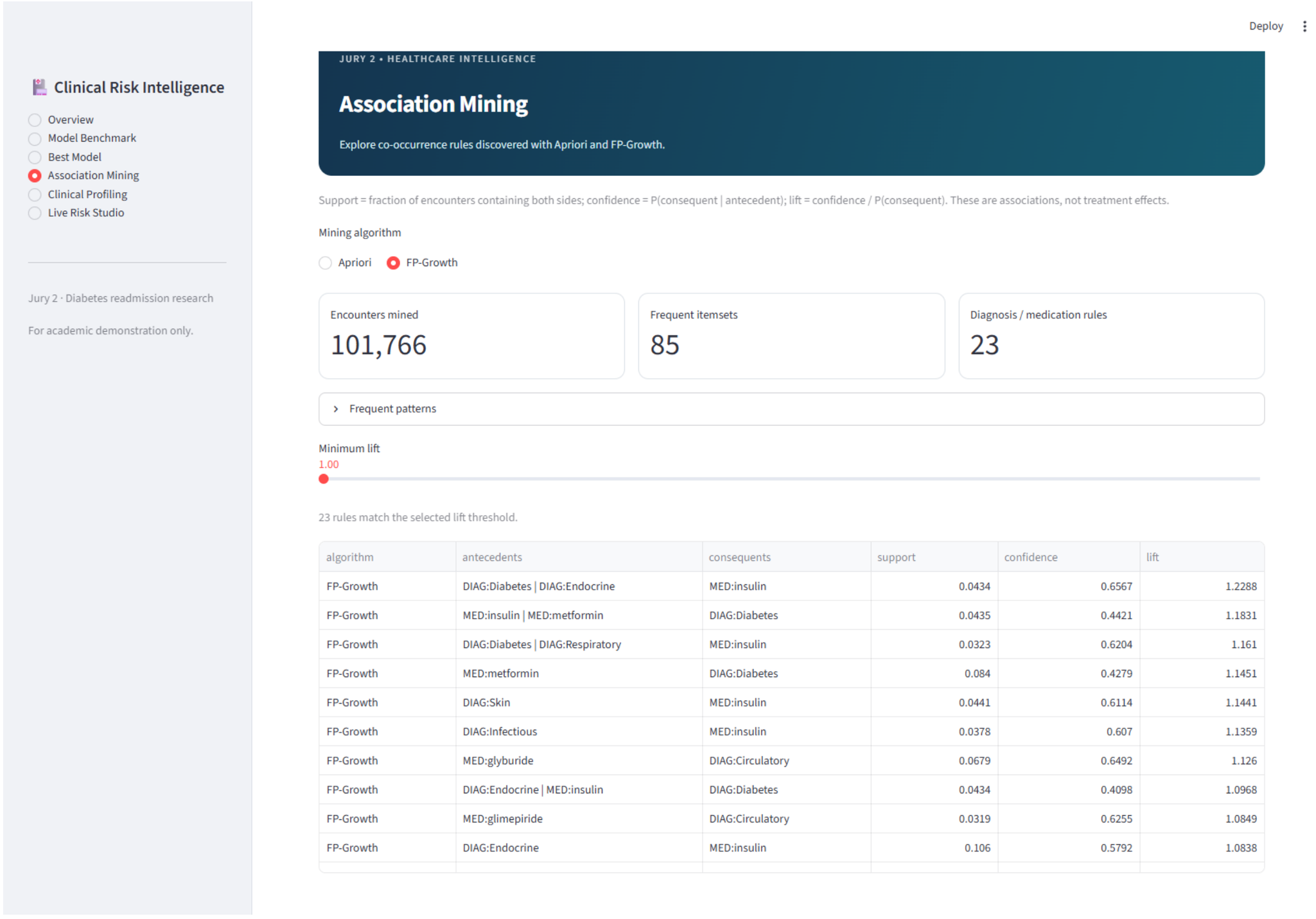
Faculty / Guide: _____

Verified implementation and evidence edition - 21 October 2026. Personal and institutional details are intentionally left blank for the team to complete.

Screenshots in this document were captured directly from the working Streamlit application in an isolated Google Chrome session. No dashboard interface or displayed result was reconstructed. Landscape A3 pages preserve screenshot readability; zoom when viewing on a smaller display.

1. Pattern Matching / Association Rule Mining

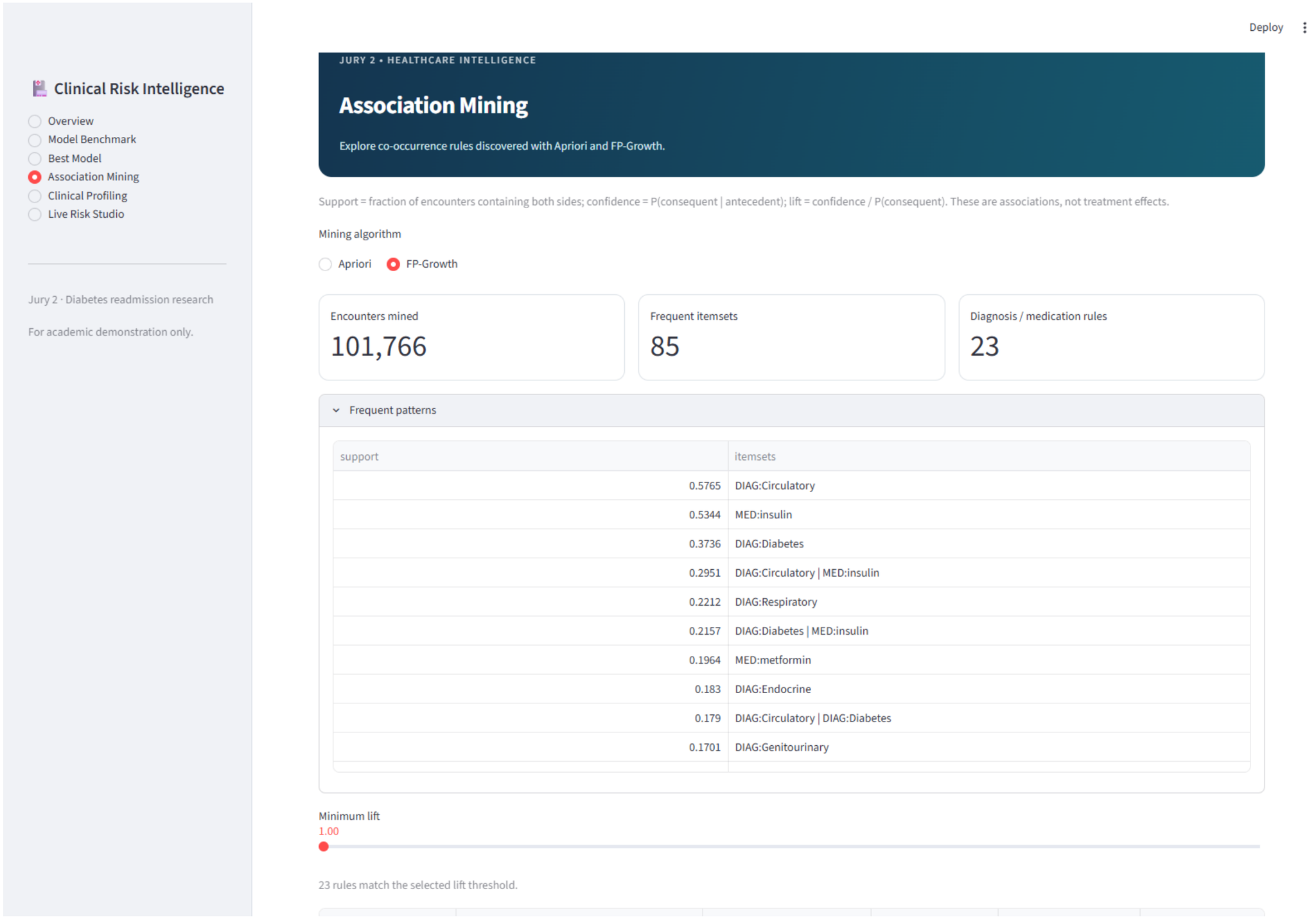
Apriori and FP-Growth were run on 101,766 encounters. Both found 85 itemsets and 23 filtered rules. The screenshot shows actual FP-Growth rule support, confidence and lift. Co-occurrence does not establish a medical treatment effect.



Genuine Streamlit viewport capture - 01_association.png

1.1 Frequent itemsets

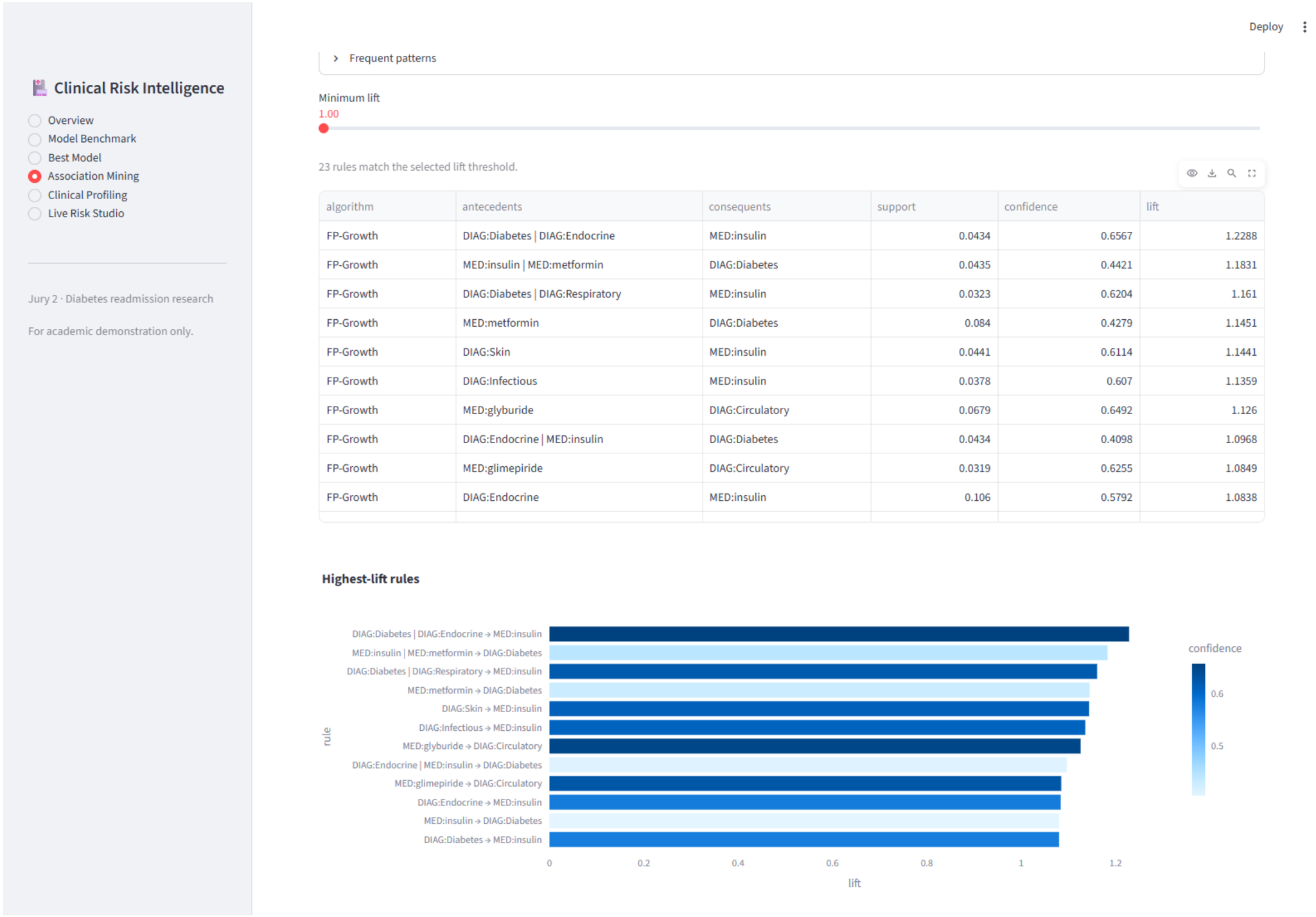
The expanded frequent-pattern table displays the itemsets underlying the rules. Thresholds: support 3%, confidence 35%, lift at least 1; maximum itemset size 3.



Genuine Streamlit viewport capture - 01_itemsets.png

1.2 Association rule chart

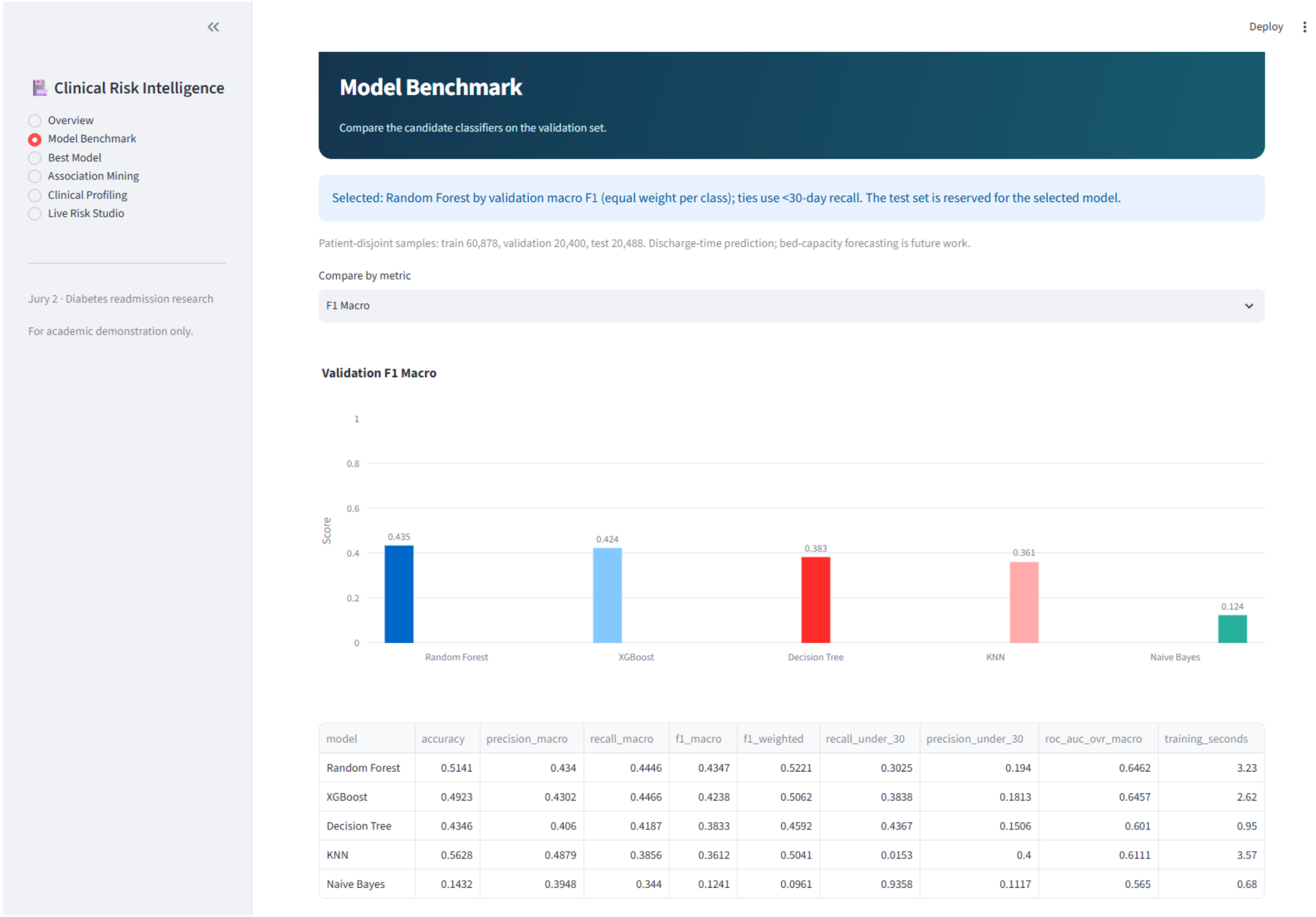
The actual dashboard chart compares lift across the highest-ranked rules, with confidence encoded by colour. The current dataset shows modest enrichment; this visual does not establish medical causation.



Genuine Streamlit viewport capture - 01_association_chart.png

2. Five-Model Classification Comparison

Decision Tree, Gaussian Naive Bayes, kNN, Random Forest and XGBoost share the same patient-separated split. Random Forest wins validation macro F1 at 0.43471; kNN has higher accuracy but almost no early-readmission recall.

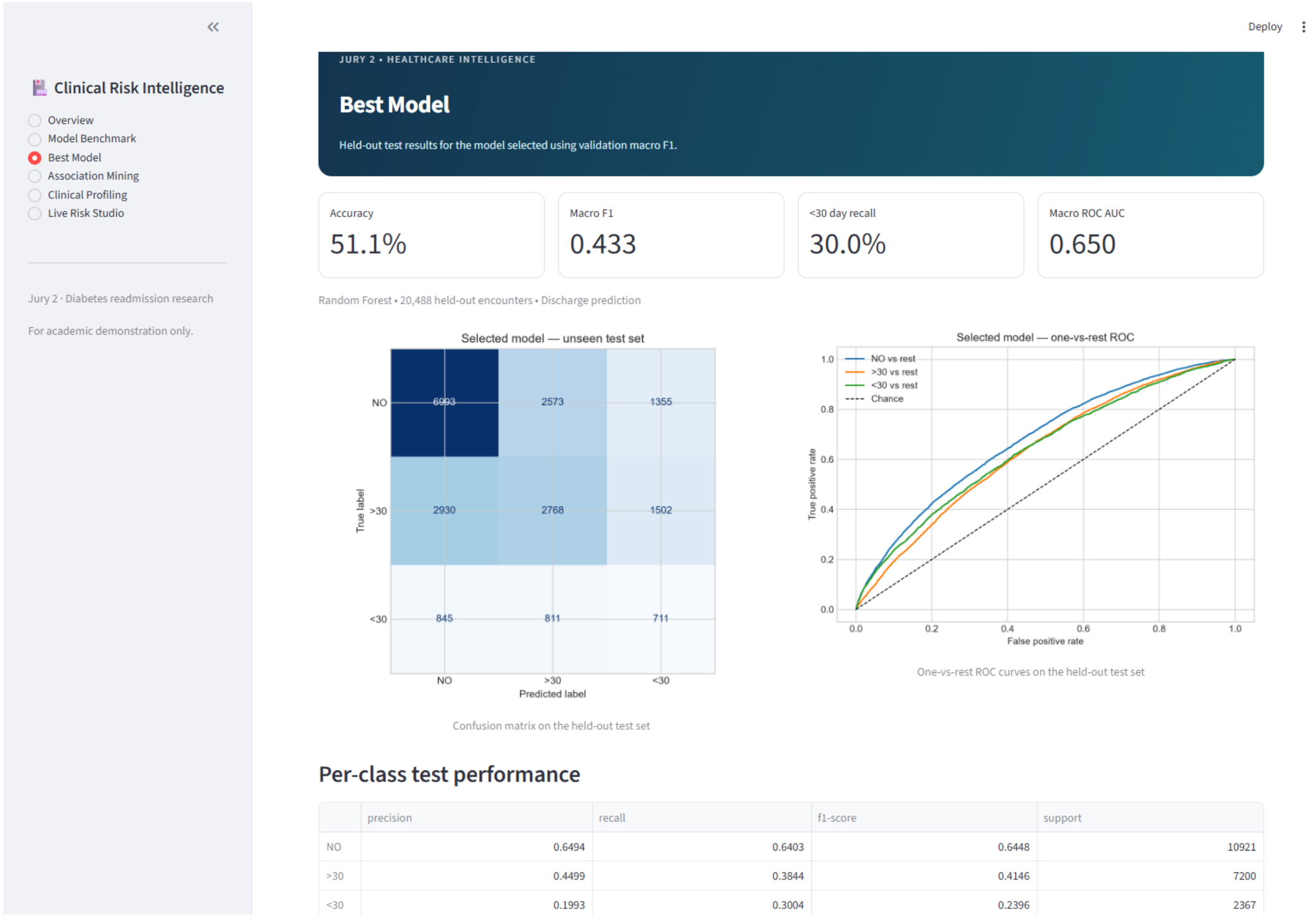


model	accuracy	precision_macro	recall_macro	f1_macro	f1_weighted	recall_under_30	precision_under_30	roc_auc_ovr_macro	training_seconds
Random Forest	0.5141	0.434	0.4446	0.4347	0.5221	0.3025	0.194	0.6462	3.23
XGBoost	0.4923	0.4302	0.4466	0.4238	0.5062	0.3838	0.1813	0.6457	2.62
Decision Tree	0.4346	0.406	0.4187	0.3833	0.4592	0.4367	0.1506	0.601	0.95
KNN	0.5628	0.4879	0.3856	0.3612	0.5041	0.0153	0.4	0.6111	3.57
Naive Bayes	0.1432	0.3948	0.344	0.1241	0.0961	0.9358	0.1117	0.565	0.68

Genuine Streamlit viewport capture - 02_benchmark.png

3. Selected Best Classifier

Random Forest on 20,488 untouched test encounters: accuracy 51.11%, macro F1 0.43302, macro one-vs-rest ROC-AUC 0.65031 and <30-day recall 30.04%. Confusion-matrix rows are true outcomes; columns are predictions.



Per-class test performance

	precision	recall	f1-score	support
NO	0.6494	0.6403	0.6448	10921
>30	0.4499	0.3844	0.4146	7200
<30	0.1993	0.3004	0.2396	2367

Genuine Streamlit viewport capture - 03_best_model.png

3.1 Classification report and interpretation

Early-readmission precision is 19.93% and F1 is 0.23960. Permutation importance is exploratory, using 250 held-out rows and two repeats. This is a discharge-time academic prototype with limited minority-class performance.

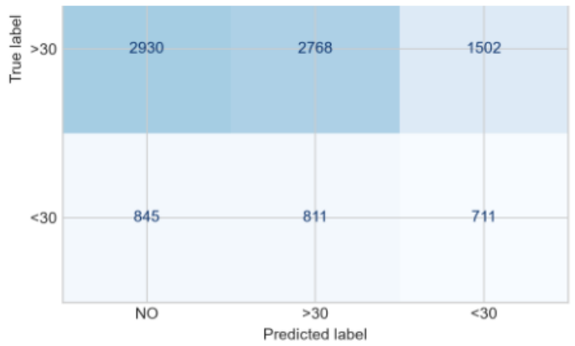
Clinical Risk Intelligence

- Overview
- Model Benchmark
- Best Model
- Association Mining
- Clinical Profiling
- Live Risk Studio

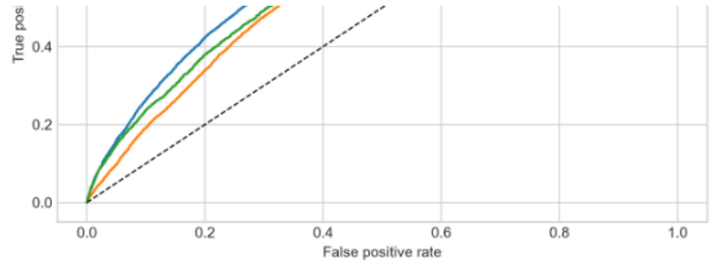
Jury 2 · Diabetes readmission research

For academic demonstration only.

Deploy



Confusion matrix on the held-out test set

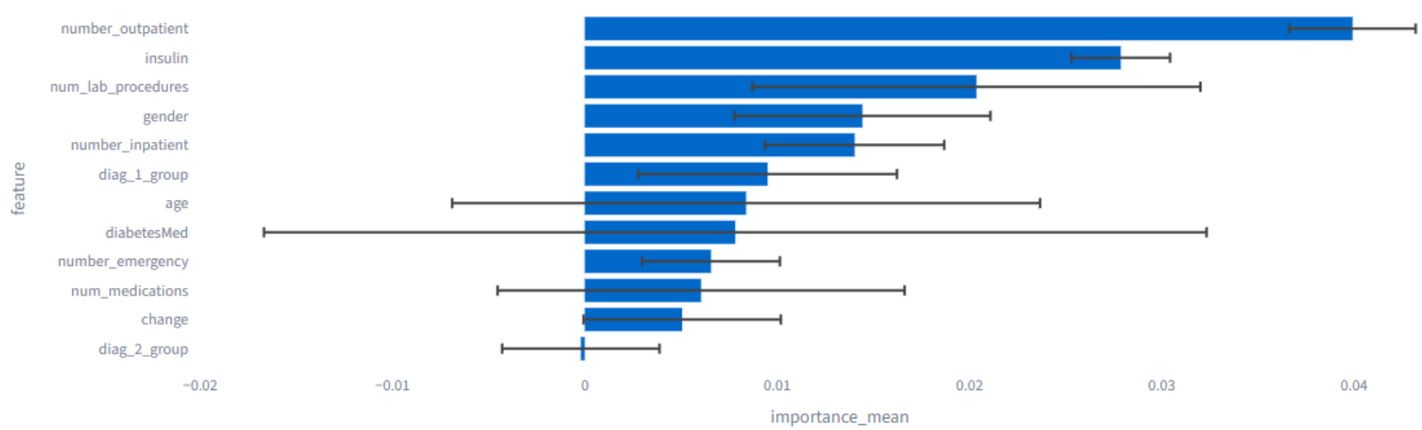


One-vs-rest ROC curves on the held-out test set

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Permutation importance (held-out sample)



Genuine Streamlit viewport capture - 03_best_details.png